

V4.2 Toy Model 2B

Two-Particle Relational Projection and Bell Correlations

Rev A - Exploratory Quantum-Side Compatibility Report

Date: 2026.05.25 | Continuation after Toy Model 2A

Status: Toy Model 2B tests whether the reciprocal projection idea can handle two-particle entanglement correlations without violating no-signalling. It is a compatibility test, not a derivation of Quantum Mechanics.

1. Purpose

Toy Model 2A showed that a normalized hidden 3D relational orientation can be mapped into a two-state spinor while preserving probability normalization and basic no-signalling compatibility. Toy Model 2B tests the next required step: a two-particle relational state.

The target is the singlet state, because it is the cleanest two-particle entanglement benchmark. A viable model must reproduce the standard singlet correlation:

$$E(a,b) = -a \cdot b = -\cos(\theta_{ab})$$

Term reminder: $E(a,b)$ is the expectation value of Alice and Bob's measurement outcomes. The dot product $a \cdot b$ measures how aligned their measurement axes are.

2. Critical first result: the naive anti-aligned local model fails

Gemini's proposed starting point was to model the hidden 3T sector as two anti-aligned orientations:

$$\begin{aligned} n_A &= n \\ n_B &= -n \end{aligned}$$

If each particle is then projected independently using the 2A rule, the conditional correlation is:

$$E(a,b | n) = -(a \cdot n)(b \cdot n)$$

If the hidden direction n is averaged uniformly, the result becomes:

$$E_{\text{naive}}(a,b) = -(a \cdot b) / 3$$

This is not the quantum singlet result. It is too weak by a factor of 3. It also cannot violate the Bell/CHSH bound. Therefore, the simple independent anti-aligned projection is rejected.

3. Surviving 2B model: non-factorizable relational projection

To reproduce singlet correlations, the hidden 3T object cannot be treated as two separately projected local orientations. It must be treated as one non-factorizable relational object.

Non-factorizable means the two-particle state cannot be split into particle A state times particle B state. This is exactly the ordinary quantum meaning of entanglement.

$$\text{Relational hidden object} \rightarrow \text{Pi}_{2B} \rightarrow |\psi_{\text{singlet}}\rangle$$

The singlet state is:

$$|\psi_{\text{singlet}}\rangle = (|01\rangle - |10\rangle) / \sqrt{2}$$

This does not derive the singlet state from first principles. It defines the surviving relational projection target needed for compatibility with known quantum physics.

4. Joint probability rule

For Alice measurement axis a and Bob measurement axis b , with outcomes s_A and s_B equal to $+1$ or -1 , the singlet joint probabilities are:

$$P(s_A, s_B | a, b) = 1/4 [1 - s_A s_B (a \cdot b)]$$

Meaning of the symbols:

- $P(s_A, s_B | a, b)$ means the probability of Alice getting outcome s_A and Bob getting outcome s_B when measuring along axes a and b .
- s_A and s_B are result labels. Each can be $+1$ or -1 .
- a and b are measurement directions.
- $a \cdot b$ equals $\cos(\theta_{ab})$, where θ_{ab} is the angle between the two axes.

- The factor 1/4 distributes probability across four possible outcome pairs.

From these probabilities:

$$\begin{aligned} P(++) &= P(--)=1/4 \quad [1 - a \cdot b] \\ P(+-) &= P(-+)=1/4 \quad [1 + a \cdot b] \end{aligned}$$

5. Expectation value check

The expectation value is calculated as:

$$E(a,b) = P(++) + P(--)-P(+-)-P(-+)$$

Substituting the probabilities gives:

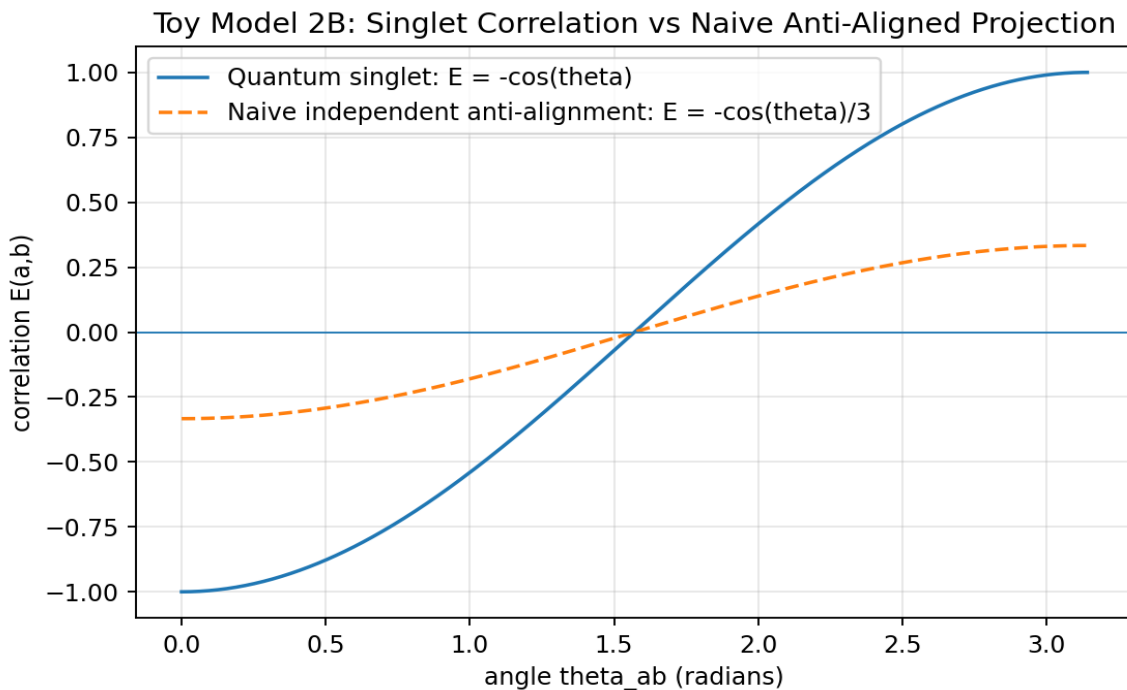
$$E(a,b) = -a \cdot b$$

This exactly matches the standard quantum singlet prediction.

6. Numerical verification

The model was tested over 10,000 random measurement-axis pairs.

Check	Result
Random axis pairs tested	10000
Max probability sum error	0.000e+00
Max probability range error	0.000e+00
Max local marginal error	0.000e+00
Max correlation error vs -a dot b	1.110e-16
Mean absolute error of rejected naive model	3.335e-01
Max absolute error of rejected naive model	6.664e-01



7. Bell/CHSH verification

The CHSH expression tests whether the model reproduces the known quantum violation of the classical Bell bound. For the chosen standard axes:

$$\begin{aligned} a &= (1,0,0) \\ a' &= (0,1,0) \\ b &= (1/\sqrt{2}, 1/\sqrt{2}, 0) \\ b' &= (1/\sqrt{2}, -1/\sqrt{2}, 0) \end{aligned}$$

The CHSH combination is:

$$S = E(a,b) + E(a,b') + E(a',b) - E(a',b')$$

Quantity	Value
$E(a,b)$	-0.707107
$E(a,b')$	-0.707107
$E(a',b)$	-0.707107
$E(a',b')$	0.707107
$ S $	2.828427
$2 \sqrt{2}$	2.828427

The model reproduces the quantum CHSH value $2 \sqrt{2}$. This confirms that the surviving 2B projection is quantum-compatible, not a local hidden-variable model.

8. No-signalling check

Even though the joint probabilities depend on both a and b , the local marginal probabilities do not.

$$P_{A(+ | a, b)} = P(++) + P(+ -) = 1/2$$

$$P_{A(- | a, b)} = P(- +) + P(- -) = 1/2$$

$$P_{B(+ | a, b)} = P(++) + P(- +) = 1/2$$

$$P_{B(- | a, b)} = P(+ -) + P(- -) = 1/2$$

Plain meaning: Alice cannot use her measurement setting to send a controllable signal to Bob, and Bob cannot use his setting to send a controllable signal to Alice.

9. Interpretation inside the 6D Projection Framework

Toy Model 2B produces an important correction to the initial intuition. Hidden 3T anti-alignment alone is not enough if it is treated as two independent local states. That version fails.

The survivable interpretation is:

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hidden 3T relational pair
-> non-factorizable projection  $\Pi_{2B}$ 
-> singlet joint amplitude
-> Bell correlations  $E(a,b) = -a \cdot b$ 
-> no controllable faster-than-light signalling
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This means the hidden 3T relation must be a relation first, not a pair of separated hidden arrows. That is compatible with the framework's principle that relational organization is prior to observer-accessible order.

10. What Toy Model 2B achieves

- It rejects the naive independent anti-aligned local projection.
- It defines the required surviving two-particle relational projection structure.
- It reproduces the singlet joint probability rule.
- It reproduces $E(a,b) = -a \cdot b$.
- It reproduces the CHSH quantum value $|S| = 2 \sqrt{2}$.
- It preserves no-signalling at the local marginal level.

11. What Toy Model 2B does not achieve

- It does not derive entanglement from first principles.
- It does not prove the physical existence of the hidden 3T sector.
- It does not provide a local hidden-variable model. Bell's theorem forbids that route.
- It does not explain measurement collapse.
- It does not yet explain decoherence.
- It does not yet generate new experimental predictions.

12. Close-out statement for 2B

Toy Model 2B establishes that the reciprocal projection architecture can be made compatible with two-particle Bell correlations only if the hidden 3T pair is treated as a non-factorizable relational object. A naive model of two independent anti-aligned hidden orientations fails, producing $E = -(a \cdot b)/3$ instead of the quantum result $E = -a \cdot b$. The surviving Π_{2B} projection reproduces singlet joint probabilities, CHSH violation, and no-signalling. This is a successful compatibility

13. Recommended next chapter

The next logical chapter is Toy Model 2C: Decoherence as Projection Stabilization.

1. Define a hidden relational superposition before measurement.
2. Couple it to an environment record field.
3. Show how interference terms become inaccessible to the 1T observer.
4. Check that the global state remains unitary while the observed subsystem appears classical.
5. Keep no-signalling and probability normalization as hard guardrails.

14. Rebind prompt for continuation

We are resuming after Toy Model 2B.

2B tested two-particle relational projection and Bell correlations.

Key result:

The naive hidden anti-aligned independent projection fails:

$$E_{\text{naive}}(a,b) = -(a \cdot b)/3$$

Surviving model:

The hidden 3T pair must be treated as a non-factorizable relational object.

Pi_2B projection target:

$$|\psi_{\text{singlet}}\rangle = (|01\rangle - |10\rangle) / \sqrt{2}$$

Joint probability:

$$P(s_A, s_B | a,b) = 1/4 [1 - s_A s_B (a \cdot b)]$$

Result:

$$E(a,b) = -a \cdot b$$

$$\text{CHSH } |S| = 2 \sqrt{2}$$

Local marginals remain 1/2, so no-signalling is preserved.

Guardrail:

This is compatibility with known quantum entanglement, not a derivation of Quantum Mechanics or proof of physical 3T existence.

Next recommended chapter:

Toy Model 2C - Decoherence as Projection Stabilization.